

Chapter 3 Complete — Gap and Research Questions

PhD-track proposal · progress review

Ali Hamed · Prof. Iris Reinhartz-Berger · Prof. Arnon

Wednesday 12 August 2026, 09:00

What you asked for on 5 August

Request (E15)	State
Finish the Gap and Research Questions chapter in full	Delivered
Per-RQ literature spreadsheet with an RQ tag column	Delivered
Think about §2 and §4 — do not start them	Thought about, not started
Word proposal + a separate tracking document	Delivered (both)
Share the Drive with Iris and Arnon	Needs Ali (Google account)
Reply to your check-in email	Awaiting the email
Present progress live today	This session

Five of seven delivered in the folder; two require Ali personally.

Your corrections — all applied, two flagged back

- E5 — "reused" removed from the main question; reuse now lives in SQ2 and SQ3
- E6 — question narrowed to variability in guideline operationalization
- E7 — "reliable" kept; auditable / transferable / end-to-end dropped from the headline
- E8 — SQ2 split into capture vs. transfer; "expert judgment" adopted
- E9 — SQ2 tied to core reasoning; evaluation criteria built into all three questions
- E12 — SQ3 rebuilt around classifying domain-specific vs. transferable uncertainty
- E13 — proposal follows the confirmed chapter structure
- Two things I could NOT resolve from the recording — I did not choose silently:
 - "exploration" vs "identification / classification" in the main question
 - "human judgment" vs "expert judgment" across the question set

The gap: four fields, each solving a different part

Body of work	Establishes	Leaves open
Automated model assessment	Deviation can be detected and scored	What a deviation MEANS
LLM / agentic assistance	Expert correction is consistently needed	WHEN to ask; WHAT to keep
Human-in-the-loop & oversight	Mechanisms and governance expectations	Selection policy under a budget
Variability engineering	Formalism for DESIGNED variability	OBSERVED interpretive variability

The gap is the CONNECTION between them: when to ask → what to keep → where it may legitimately be reused.

Openness statements are candidate claims — the systematic searches are frozen and not yet run.

One question, three sub-questions

U-RQ · How can human judgment be captured, governed, and used to support agentic-AI-driven variability exploration in guideline operationalization scenarios, enabling reliable human–AI co-reasoning?

	Asks	Built-in evaluation criterion
SQ1	WHEN should the system ask an expert?	Coverage of important uncertainty AND bounded burden
SQ2	WHAT must be kept, under what governance?	No unsafe generalization; no loss of human authority
SQ3	WHERE ELSE does a judgment hold?	Classification must discriminate; transfer must survive leakage controls

Each question can fail on its own terms — and a negative result is still a reportable contribution.

Literature: 40 sources tagged — and what they do NOT cover

- 40 verified sources, each tagged RQ1 / RQ2 / RQ3 / general, independently fabrication-checked
- Coverage-gap analysis per question — the part beyond an inventory:
 - RQ1 — nothing on interrupt/approval mechanics inside a multi-agent LLM pipeline
 - RQ2 — weakest coverage: no agent-memory or learning-from-feedback sources
 - RQ3 — no usability instruments, no cross-domain transfer studies
 - general — NO design-science methodology sources, though the method is design science
- Frozen search protocol QL-01...QL-05 ready to execute on your word

Evidence position — unchanged, and stated plainly

Gate	Value	What it blocks
Independent expert labels (EXP-005)	0 of 24	Every quantitative accuracy claim (≥ 20 needed)
Medical entry gates G1–G6	0 of 6	Any medical data processing
Literature searches QL-01...QL-05	Not run	Any novelty or completeness statement
Accuracy / generalization / clinical claims	None made	— checked by an automated guard

The binding constraint is not writing or engineering — it is independent expert labels. That needs two reviewers and an adjudicator: a resourcing decision, not a build task.

§2 and §4 — thought about, deliberately not started

- §2 literature survey: four structural options with trade-offs (SQ-parallel / community-parallel / lifecycle-stage / hybrid) — none selected
- §4 research artifact: nine options across SQ1–SQ3, at different abstraction levels — none selected
- A recurring issue surfaced: the current "core artifact" rows bundle 6, 9 and 10 components; several SQ3 items are study OUTPUTS, not artifacts
- 14 open questions prepared for you — grouped by chapter shape, artifact definition, evidence admissibility, and resourcing

Eight decisions I need from you

ID	Decision	Blocks
D-RQ-01	Final U-RQ and SQ1–SQ3 wording (incl. the two flagged discrepancies)	Everything downstream
D-RQ-02	Ratify or reverse dropping auditable / transferable / end-to-end	Title and headline
D-TITLE-01	Proposal title — three candidates offered	Submission-readiness
D-DECOMP-01	Are three sub-questions jointly sufficient for "reliable co-reasoning"?	Chapter 3 sign-off
D-CH2-01	Chapter 2 structure — which of four options	Starting §2
D-ART-01	One named artifact per study, or a package?	Starting §4
D-SEARCH-01	Run searches before or after wording sign-off? Add QL-06+?	Literature execution
D-PRELIM-01	Which preliminary results may appear, labelled how?	Chapter 5

The first three block everything else.

Where this leaves us

- Chapter 3 is complete, cited, and honest about what is still a candidate claim
- The literature map exists — and its gaps are now visible rather than latent
- §2 and §4 are ready to start the moment you confirm direction and wording
- Two wording questions and eight decisions are on the table today
- Next after this meeting: apply your rulings in one pass, then execute the searches